That's actually a **solid and well-structured plan** — you're aligning your interests (AI/ML) with career readiness (DSA, aptitude, soft skills, resume building).
Let me break it down into a **practical roadmap** so you can execute it smoothly without burning out.

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1. AI/ML Learning Path (Parallel to Your Degree)

Since you're avoiding IoT, you can fully focus on AI/ML skills.

Phase 1 — Fundamentals (2–3 months)

- **Python** (You already like it, but focus on AI-relevant areas):
  - Numpy, Pandas, Matplotlib/Seaborn (data handling + visualization)
  - Data preprocessing basics
- **Math for ML**:
  - Linear algebra basics (vectors, matrices, dot products)
  - Probability & statistics (mean, variance, distributions)
  - Basic calculus (derivatives, gradients)
- **Intro to ML** (Scikit-learn):
  - Supervised vs. Unsupervised learning
  - Simple models: Linear Regression, Decision Trees, KNN, Naive Bayes
  - Practice with small datasets (Iris, Titanic, MNIST subset)

Phase 2 — Deep Learning & AI Applications (3–4 months)

- **Neural Networks** (PyTorch or TensorFlow)
- Image-based AI (CNNs) → projects like image classification
- Text-based AI (NLP, Transformers) → projects like sentiment analysis or chatbot
- Work on **mini-projects** for GitHub:
  - Spam Email Classifier
  - Handwritten Digit Recognition
  - Movie Recommendation System
  - AI Chatbot

Phase 3 — Resume-Ready AI Projects (Ongoing)

- Pick **real-world datasets** from Kaggle & build end-to-end projects.
- Host them on GitHub with **clean README + deployment** (Streamlit, Hugging Face Spaces).

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2. DSA & Aptitude (Interview Readiness)

DSA Roadmap (1–1.5 years alongside AI/ML learning)

- Arrays, Strings, Linked Lists, Stacks, Queues, Recursion
- Searching & Sorting algorithms
- Trees & Graphs
- Dynamic Programming (later stage)
- Practice daily: **LeetCode, GeeksforGeeks, Codeforces**

Aptitude Skills (for placements)

- Quantitative: percentages, ratios, probability, speed-distance, permutations
- Logical Reasoning: puzzles, sequences, data interpretation
- Resources: IndiaBix, PrepInsta

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3. Soft Skills & Resume Building

- Public speaking practice (college events, mock interviews) to reduce nervousness
- Resume tips:
    - Highlight AI/ML projects + GitHub
    - Add certifications (Coursera, Kaggle, Google AI)
    - Keep it **1 page, achievement-focused**

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4. Suggested Weekly Schedule

Mon–Fri (2–3 hrs/day)

- 1 hr AI/ML
- 1 hr DSA or aptitude
- 30 min soft skills or resume work

Sat–Sun (4–5 hrs)

- Deep dive into projects, coding contests, or AI learning

 By the time you're in **final year**, you'll have:

- **3–5 solid AI projects on GitHub**
- **Strong DSA problem-solving skills**
- **Polished resume + placement-ready aptitude**

- A niche skill (AI/ML) that differentiates you from the IoT-heavy crowd

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If you want, I can make you a **month-by-month AIML + DSA + placement prep timeline** so you know exactly what to learn each month. That way, there's no confusion on what to focus on next.

Got it — let's focus specifically on **AI/ML** so you know exactly **what topics, tools, and concepts** to learn from start to finish.

I'll arrange this **step-by-step** so you can go from zero → project-ready → placement-ready in AI/ML.

□ AI/ML Learning Path for You

1 □ Foundations (Month 1–2)

Goal: Build the programming + math base AI needs.

- **Python for Data Science & AI**
  - Data types, loops, functions, OOP
  - Libraries: **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**
- **Math for ML**
  - **Linear Algebra**: vectors, matrices, dot product, transpose
  - **Statistics**: mean, median, variance, probability distributions
  - **Calculus basics**: derivatives, gradients (only basics needed initially)

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2 □ Machine Learning Basics (Month 3–4)

Goal: Understand core ML algorithms + implement them.

- **ML Workflow**: data collection → preprocessing → model training → evaluation
- **Supervised Learning**
  - Linear Regression, Logistic Regression
  - Decision Trees, Random Forest, KNN, Naive Bayes, SVM
- **Unsupervised Learning**
  - K-Means, Hierarchical Clustering, PCA (dimensionality reduction)
- **Model Evaluation**
  - Accuracy, Precision, Recall, F1-score, Confusion Matrix
- **Tools**: **Scikit-learn**, Jupyter Notebook
- **Mini Projects**:

    - Titanic survival prediction
    - Iris flower classification
    - Movie recommendation system (basic)

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3 Deep Learning (Month 5–6)

Goal: Learn Neural Networks & handle image/text data.

- **Neural Networks (NNs)**
    - Perceptron, Activation functions (ReLU, Sigmoid, Softmax)
    - Forward & backward propagation
    - Loss functions & optimizers
- **Image Processing (CNNs)**
    - Convolution, pooling, filters
    - Architectures: LeNet, VGG, ResNet basics
- **Text Processing (RNNs, LSTMs)**
    - Word embeddings, Tokenization
    - Sequence modeling basics
- **Tools**: **TensorFlow/Keras** or **PyTorch** (choose one)
- **Projects**:
    - Handwritten digit recognition (MNIST)
    - Dog vs Cat classifier
    - Sentiment analysis on tweets

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4 Advanced AI Topics (Month 7–8)

Goal: Work on industry-level AI use cases.

- **Natural Language Processing (NLP)**
    - Transformers (BERT, GPT basics)
    - Text summarization, Named Entity Recognition
- **Transfer Learning**
    - Using pre-trained models for custom datasets
- **Reinforcement Learning (Intro)**
    - Agent, environment, reward system basics
- **Projects**:
    - AI Chatbot (Rasa or simple LLM-based)
    - Fake news detection
    - Resume screening AI

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5 Model Deployment & Real-World Integration (Month 9–10)

Goal: Showcase your work professionally.

- **Model Deployment**
  - Streamlit, Flask, FastAPI for web apps
  - Deploy on Hugging Face Spaces, Heroku, or AWS
- **Version Control**: Git & GitHub for code sharing
- **Experiment Tracking**: MLflow or Weights & Biases (optional)
- **Projects**:
  - AI-powered website or app
  - Dashboard with AI insights

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6 Resume-Ready AI/ML Profile (Month 11+)

Goal: Stand out in placements.

- **Portfolio Projects (4–6)** — mix of image, text, and tabular data
- **Certifications**:
  - Google AI/ML Crash Course
  - Coursera's Machine Learning by Andrew Ng
  - Kaggle Competitions participation
- **GitHub Profile**: Each project should have
  - README with explanation
  - Dataset source
  - How to run the project

 Pro Tip: Since you also want to do **DSA + Aptitude** for placements, you can spend **1–2 hrs/day on AI/ML** while keeping **1 hr/day for DSA** in parallel.

If you want, I can make you a **month-by-month AI/ML + project schedule** so you'll know *exactly* what to learn, when, and which project to build at each stage — no guesswork.
That would give you a complete 1-year AI/ML mastery plan.